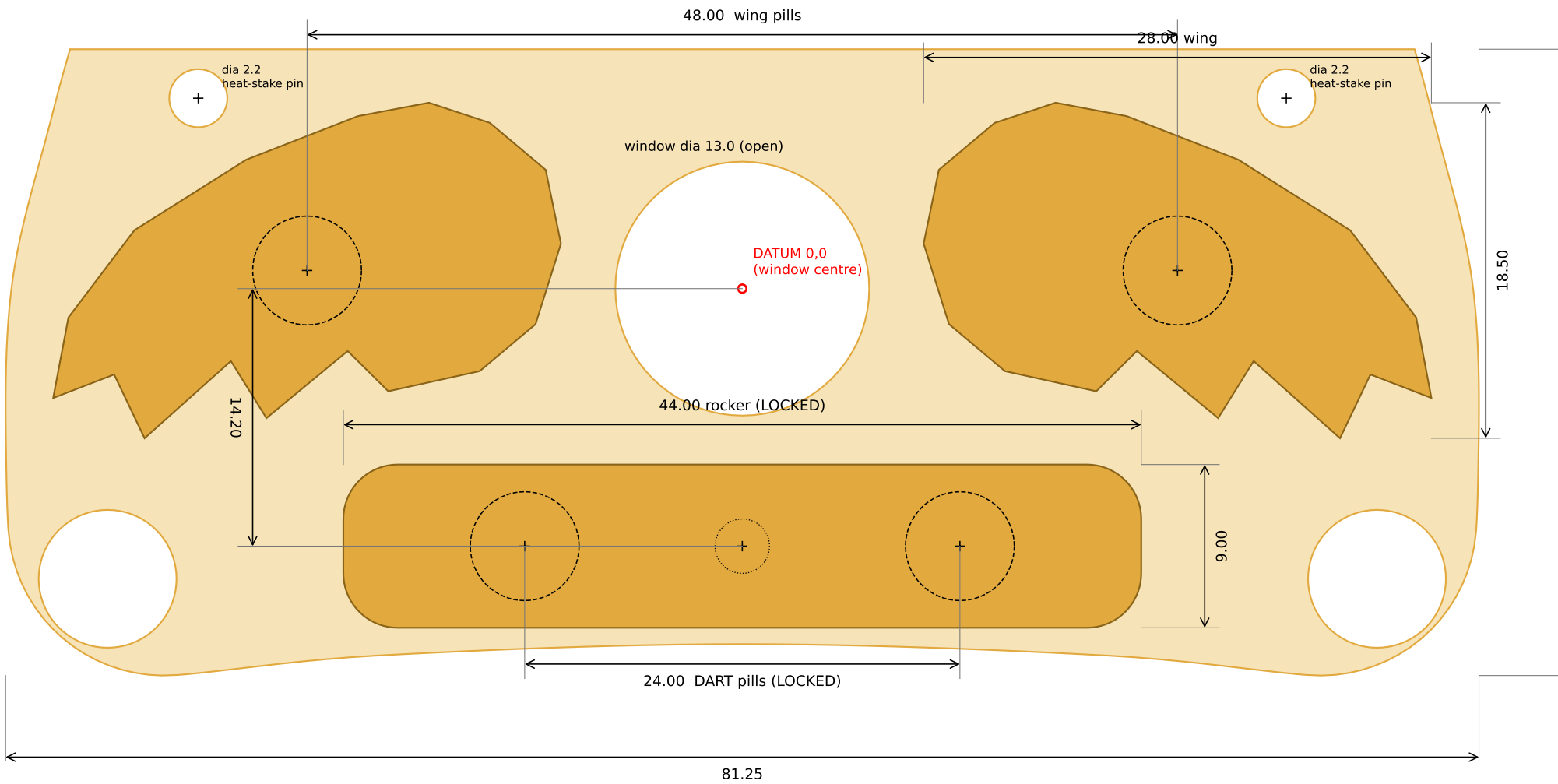


STRUTHIO-CM1 conductive-silicone control mat

one-piece compression-moulded keypad: LEFT WING, RIGHT WING, two-end DART rocker; 4 carbon pills

TOP VIEW (player side) - key tops gold, web pale, pills dashed



MATERIAL / FINISH

VMQ silicone rubber, 50 Shore A nominal (+5 for first-sample tuning only)

Colour: STRUTHIO gold, ref. #E2A93F (approve on first samples)

Key tops matte; no printing

CONTACTS

4 conductive carbon pills, dia 6.0 x ~0.5, flush in the plunger faces

Contact resistance <= 100 ohm (on ENIG interdigitated pads, STRUTHIO-CP1)

Pill centres: wings (+-24.00, +1.00); DART (+-12.00, -14.20): 24.00 apart (LOCKED)

FEEL TARGETS (LOCKED)

WINGS: travel 1.5, peak actuation 125 g, snap ratio 45-55 %

DART ENDS: travel 1.3, peak actuation 150 g, snap ratio 40-50 %

Rocker pivots on the opposite end flange; centre stop nub ends 1.10

above the PCB: a flat press stops before either pill (no double dart)

Skirt geometry is the moulder's, to meet the targets: report force-

travel curves for 5 keys of the first samples

HEIGHTS (from the PCB face = mat underside)

Web thickness 1.0; web lies flat on the PCB

Wing key top 7.4; rocker top 7.1

Key flange 2.0 to 2.6 (captured behind the 3.0 shell face)

Wing pill face 1.5; DART pill faces 1.3; centre nub 1.1

Shell holes = key outline + 0.30; flange = outline + 1.00

FILES

dxg/keys_outline.dxf (key tops), dxg/mat_outline.dxf (web)

stl/struthio_a15_mat.stl (3D reference, handheld coordinates)

Handheld coordinates = drawing + (0, -47.0)

INSPECTION

First article: outline +-0.10 on key tops, +-0.15 elsewhere; pill position +-0.15

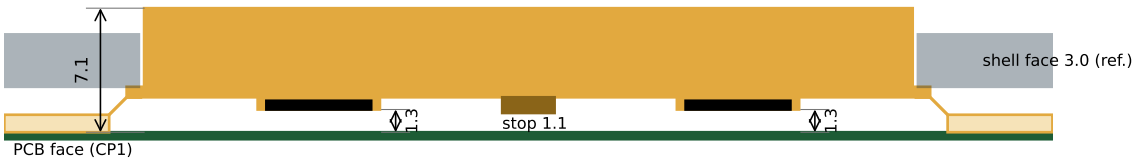
Force / travel / snap per key; contact resistance per pill; 10 x full-face press:

no sticking, rebound < 100 ms

SECTION A-A wing key through its pill (true scale)



SECTION B-B DART rocker through both pills + centre stop (true scale)



PART	STRUTHIO-CM1 rev A1.5
PROJECT	STRUTHIO handheld, A1.5 (Build Manual 1.5 lock)
SOURCE	cad/a15/STRUTHIO15.scad, checked by check_a15.py
UNITS	mm; datum = speaker-window centre; general tol. +-0.15
OWNER	R.A. Peddycoart